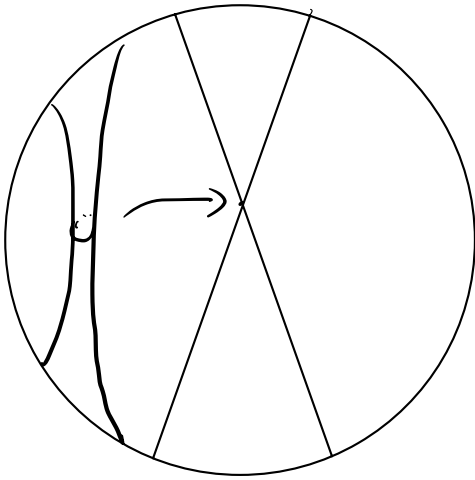


$f: Y \rightarrow T$ hol. x isol. singularity of f at x .

$$Y \rightarrow \Lambda$$



$$Y - Y_x \rightarrow T - t_0$$

$$\exists \pi_1(T - t_0) \hookrightarrow H^1(Y_t, \mathbb{Z})$$

$$\sim h_{f,x} : H^1(Y_t, \mathbb{Z}) \rightarrow H^1(Y_A, \mathbb{Z})$$

Local P.L. Monodromy.

$$\Delta_{f,x}(1) = \det(1 - h_{f,x})$$

* move from integer coefficients to complex coefficients

• Integr. of PL-monodromy as monodromy of a connectn

$\hookrightarrow$ Singular Gauss Manin connectn on Cohom B .
regular.

locally trivial fibration

$f: Y \rightarrow T$ s.t. $\forall p \in T. \exists U \ni p$ w/

i) $\exists \phi: f^{-1}(U) \rightarrow U \times f^{-1}(p)$ diffeo

ii) $f^{-1}(U) \rightarrow U \times f^{-1}(p)$

$\searrow_U \swarrow$

Stresmann: proper submersion is locally trivial.

Thm (BC)

$$\begin{array}{ccc} Y \times X & \xrightarrow{g_1} & Y \\ f_1 \downarrow & & \downarrow f \\ X & \xrightarrow{g} & T \end{array}$$

then $g^* R^1 f_* \mathcal{F} \simeq R^1 f_* g^* \mathcal{F}$

Cor $(R^1 f_* \mathcal{F})_t \cong H^1(Y_t, \mathcal{F}|_{Y_t})$

Thm Y non-singular

there exists:

$$H^*(\Omega_Y \cdot dR) \xrightarrow{\sim} H_{dR}^*(Y^{an}) \cong H^*(Y^{an}; \mathbb{C})$$

Cor $f: Y \rightarrow T$ proper / a

$\exists$ natural isom. $\underbrace{(R^1 f_* \Omega_Y \cdot dR)^{an}}_{\text{analytic}} \simeq \underbrace{R^1 f_*^{an} \in \otimes \mathcal{O}_T^{an}}_{\text{algebraic}}$

GM: Recover $R^i f_* \mathbb{C}$ from $R^i f_* \mathbb{C} \otimes_{\mathbb{C}_T} \mathcal{O}_T$

Define $\nabla : R^i f_* \Omega^j \rightarrow \Omega^j \otimes_{\mathbb{C}_T} \Omega_T^1$

$\leadsto$ sheaf of flat sections $\ker \nabla = R^i f_* \mathbb{C}$
 ∇ preserves the integral lattice

Picard Fuchs How do the embeddings of

$$R^0 f_* \Omega^1 \gamma|_T \hookrightarrow R^1 f_* \Omega^1 \gamma|_T$$

vary w.r.t. to ∇^{GM}

$$f : Y \rightarrow T \quad Y_t \hookrightarrow f^{-1}(U)$$

induces $H^k(Y_t, \mathbb{C}) \cong H^k(f^{-1}(U), \mathbb{C})$

$\leadsto$ local trivialization

$\leadsto \nabla^{GM}$ is flat $\Rightarrow$ transition functions are const.

$$H^n \rightarrow T \quad \text{Bundle w/ fibres } H^n(Y_t, \mathbb{C})_{t \in T}$$

$$H^n \rightarrow T \quad \text{for } R^n f_* \mathbb{C}_X \otimes_{\mathbb{C}_T} \mathcal{O}_T$$

$$\nabla_{\partial t} : H^n \rightarrow H^n$$

$$\log \mapsto \mathbb{C} \otimes \frac{dg}{dt}$$

$$V_T \xrightarrow{f} T$$

$$V_{\Lambda}' \xrightarrow{F} \Lambda' = \Lambda - \Sigma$$

$$H_f \quad H_F$$

Period Map. ω hol. n -form on V_{Λ}'

$$\leadsto [w] = [w_1] \in H^{n-1}(V_{\Lambda}, \mathbb{C})$$

$$\int w_1 \leadsto \text{depends holomorphic on } I$$

$\partial a \rightarrow \mathbb{Z}$ -cycle

Example $f = z_1^2 + \dots + z_n^2$

$$V_{T'} \longrightarrow T' \quad S_t = \sqrt{t} S^{n-1}$$

$$\langle w_t, S_t \rangle_t = \int_{S_t} w_t = \int_{B_t} dw_t = c t^{n/2}$$

Residue form

ω a n -form on $\mathbb{C}^n$

$$f: \mathbb{C}^n \rightarrow \mathbb{C}$$

V_t - non-singular.

Then $\exists$ $(n-1)$ form ψ defined in a nbhd of V_t
s.t. $\omega = df \wedge \psi$ [$\psi|_{V_t} = \frac{\omega}{df_t} \leadsto$ Residue
form of ω at t]

$\leadsto$ Define a holomorphic section $[\frac{\omega}{df}] : T' \rightarrow \mathcal{H}_f^*$

we call such sections geometry sections.

$\underbrace{\alpha_1(\lambda), \dots, \alpha_n(\lambda)}_{\text{basis of } \mathcal{H}_{n-1}(V_\lambda, \mathbb{Z})}$ continuous w.r.t. $\lambda \in \Lambda'$

take $\omega_1, \dots, \omega_n$ n -forms on V_Λ

$$\Delta : \lambda \mapsto \det^2 \left[\int_{\alpha(\lambda)_j} \left(\frac{\omega_i}{dF} \right) \right]_{ij}$$

residue form
in the Milnor fibration

det Jacobian of γ_{GM}

Thm $\exists w_1, \dots, w_n$ s.t. $\Delta \neq 0$ on Λ'

$$H^*(V\Lambda'/\Lambda') \sim H^*(V\Lambda/\Lambda) \rightarrow \text{coherent sheaf.}$$

Fact Δ is simple valued.